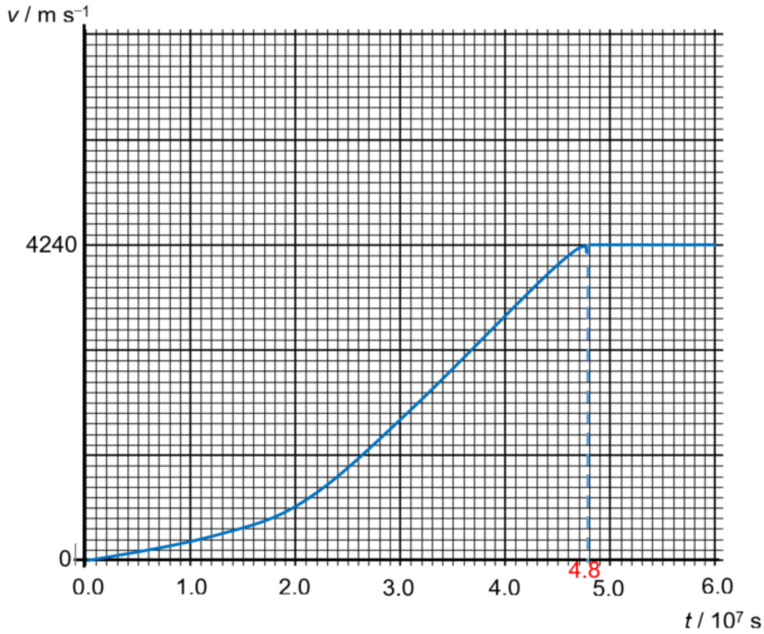
Q2	
(a)(i)	Constant velocity in the horizontal direction and a constant acceleration in the vertical direction
(a)(ii)	$v^2 = u^2 + 2as$
	$0 = (20.0 \sin\theta)^2 + 2(-9.81)(15.8)$
	$\theta = 61.7°$
	$\theta = 62°$ (0 d.p.) (shown)
(a)(iii)	$v_y = u_y + a_y t$ $0 = 20.0 \sin 61.7° - 9.81t$
	t = 1.80 s (3 s.f.)
(b)(i)	Since force exerted on the spaceship F is constant but mass of the spaceship decreases with time, given that $F = ma$, a increases with time.
(b)(ii)	$F_{\text{on fuel}} = v_{rel} \dfrac{dm}{dt}$ $= (3.0 \times 10^4)(1.7 \times 10^{-6})$
	= 0.051 N $\lvert F_{\text{on spaceship}} \rvert = \lvert F_{\text{on fuel}} \rvert$
(b)(iii)	Change in velocity = area under graph = ½(9.450 + 8.200)$\times10^{-5} \times 4.80\times10^{7}$
	= 4240 m s^{-1}
	Final velocity = 4240 – 0 = 4240 m s^{-1}

(b)(iv)



Increasing gradient from t = 0 to t = 4.8 × 10^7 s, starting from v = 0 to final v at 4240 m s^{-1}.

Between t = 4.8 × 10^7 to t = 6.0 × 10^7 s, there is no more force. Therefore, constant v at 4240 m s^{-1}.

Q3